

ECON100B - SECTION 4

The Cobb Douglas Growth Model

Felix Zhang

ANNOUNCEMENTS/AGENDA

Announcements

- PSET 1 due Friday, 2/13

Agenda

- The Cobb Douglas Model
- PSET problems

Question of the Day

1. PSET 1, Q1a: Recall that GDP measured on the expenditure side is given by $y = C + I + G + NX$. Roughly how big are each of these components as a percentage of GDP in the U.S. today?
(Use slide deck for Class 2 if needed!)
2. What is the best pizza topping? What is the worst?

The Cobb Douglas Model





A simple model

Our economy has one good we're producing: pizza

- We have a fixed number of people (our Labor), and fixed number of pizza ovens (our Capital)
- Number of workers / "labor" = $\overline{L}$
- Number of pizza ovens / "capital" = $\overline{K}$
- We include a bar here to denote fixed and exogenous (given) variables
- Firms decide how much labor to hire and how many pizza ovens to rent

A simple model

Enter the Cobb Douglas production function:

- Tells us how much pizza / output (Y) can be produced if L workers and combined with K machines:

$$Y = F(K, L) = \overline{(A)} K^{\alpha} L^{1-\alpha}$$

$\overline{(A)}$

represents some productivity parameter that is a given exogenous variable

- In our base model, K and L don't have bars because our model describes how any amounts of capital and labor can be combined to generate output, so we can scale and change K and L

Constant Returns to Scale

- Cobb Douglas exhibits constant returns to scale: For example, if we double the amount of each input, we double the amount of the output

How do we know Cobb-Douglas has the CRS property?

$$\begin{aligned}F(2K, 2L) &= A (2K)^{\alpha}(2L)^{1-\alpha} \\&= A 2^{\alpha}K^{\alpha}2^{1-\alpha}L^{1-\alpha} \\&= A (2^{\alpha}2^{1-\alpha})(K^{\alpha}L^{1-\alpha}) \\&= A (2^{\alpha + 1 - \alpha})(K^{\alpha}L^{1-\alpha}) \\&= A 2 (K^{\alpha}L^{1-\alpha}) \\&= 2F(K, L)\end{aligned}$$

Constant Returns to Scale

- Cobb Douglas exhibits constant returns to scale: For example, if we double the amount of each input, we double the amount of the output
- The fact that the exponents sum to 1 delivers constant returns to scale. If the exponents summed to more than 1, we would have increasing returns to scale, and if the exponent summed to less than 1, we would have decreasing returns to scale
- How to think about constant returns to scale: If we hired twice as many of the exact same workers and had twice as many of the exact same pizza ovens, we would produce twice as many pizzas
- The standard replication argument

PRACTICE: RETURNS TO SCALE

1. Consider $F(K, L) = K^{0.5}L^{0.5}$. What happens to output when capital triples and labor triples?
2. Consider $F(K, L) = K^{0.5}L^{0.5}$. What happens to output when capital increases by 50% and labor stays constant?
3. If $F(K, L) = K^2 + L$, does this production function exhibit increasing, constant, or decreasing returns to scale?

PRACTICE: RETURNS TO SCALE

1. **Y triples, due to constant returns to scale**

$$2. F(1.5K, L) = (1.5K)^{0.5}L^{0.5} = (1.5K)^{0.5}L^{0.5} \approx 1.225(K^{0.5}L^{0.5}) = 1.225F(K, L) \Rightarrow \text{Y increases by 22.5\%}.$$

$$3. \text{ For } z > 1: F(zK, zL) = (zK)^2 + zL = z^2K^2 + zL > zK^2 + zL = zF(K, L)$$

Increasing returns to scale.



Allocating Resources

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- Firms will want to maximize profits: π

Allocating Resources

- So how do we decide how much labor and how much capital to use?
- Firms will want to maximize profits: π
- Firms will then want to maximize the following problem (microeconomics/math review):

$$\max_{K,L} \Pi = F(K,L) - rK - wL,$$

- With capital have a “rental price” r and labor having a “wage price” w
- All our variables are denoted in some sort of variable – in our case pizza, but can be money, other goods, etc.



Diminishing Returns

Let's go back to our base models, with the following parameters

$$Y = F(K, L) = \overline{A} K^{1/3} L^{2/3}$$

- We often use 1/3 and 2/3 in a base Cobb-Douglas model due to empirical evidence of 1/3 of payments going to capital and 2/3 to labor

This also tells $wL + rK = Y$

Which also tells us that total income in the economy equals total production!



Diminishing Returns

Let's go back to our base models, with the following parameters

$$Y = F(K, L) = \overline{A} K^{1/3} L^{2/3}$$

- Holding Labor, L constant, what happens when we increase Capital?

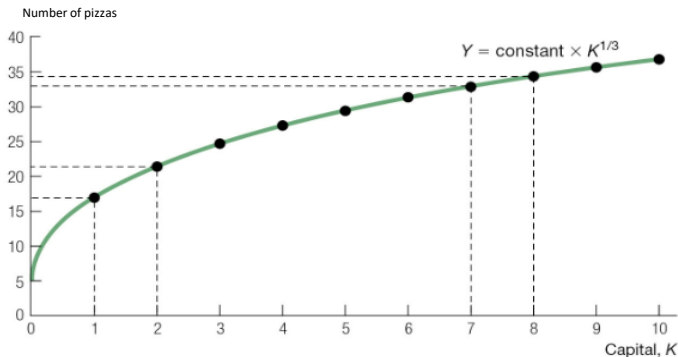


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Marginal Product of Capital (MPK)

MPK is the additional output/return from one more unit of capital:

$$MPK := \frac{\partial F(K, L)}{\partial K}$$

For example, with Cobb-Douglas:

$$MPK = \alpha K^{\alpha-1} L^{1-\alpha} = \alpha \frac{Y}{K}$$

or the product of the factor's exponent and the average amount that each unit of the factor produces

Marginal Product of Labor (MPL)

MPL is the additional output/return from one more unit of labor:

$$MPL := \frac{\partial F(K, L)}{\partial L}$$

For example, with Cobb-Douglas:

$$MPL = (1 - \alpha)K^{\alpha}L^{-\alpha} = (1 - \alpha) \frac{Y}{L}$$

- For capital, we “hire” more capital until $MPK = r$
- For labor, we hire more labor until $MPL = w$

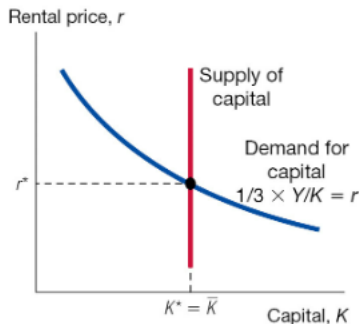
Solving the equation

Solving for our capital and labor mix gives us our general equilibrium:

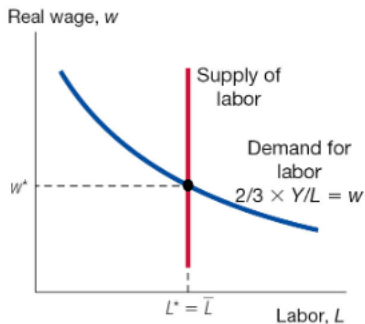
- Clearing of our capital market
- Clearing of our labor market
- Modeling the entire economy!

Solving the equation

Assuming we are given a certain supply of capital and certain supply of labor:



(a) The capital market



(b) The labor market

PRACTICE: MPK, MPL, AND LABOR DEMAND

1. Suppose the production function is $F(K, L) = 10K + 2L$. If the firm currently hires 4 workers, calculate the MPL.
2. If the firm is deciding between hiring one more worker or renting an additional machine and $W = R$, which one should it choose if $MPL = 5$ and $MPK = 10$?
3. Calculate the change in output if labor increases from 5 to 6, given $MPL = 8$.

PRACTICE: MPK, MPL, AND LABOR DEMAND

1. $MPL := \frac{\partial F(K,L)}{\partial L} = 2$ so doesn't depend on $L \Rightarrow \mathbf{MPL = 2}$
2. **The firm should rent more capital since marginal costs are identical and $MPK > MPL$.**
3. **Since MPL is how much Y changes with a one unit increase in L , the output increases by 8 units.**

Output per person

- Cobb Douglas tells us about total output, but not output per person
- Per capita = per person, per worker = per member of labor force

- Output per person: $y = Y/L$
- Capital per person: $k = K/L$

- Output per person is higher when:
- The productivity parameter is higher
- The amount of capital per person is higher

$$\begin{aligned} y^* &\equiv \frac{Y^*}{L^*} = \frac{\bar{A} \bar{K}^{-1/3} \bar{L}^{2/3}}{\bar{L}} \\ &= \frac{\bar{A} \bar{K}^{-1/3}}{\bar{L}^{1/3}} \\ &= \bar{A} \bar{k}^{-1/3}. \end{aligned}$$

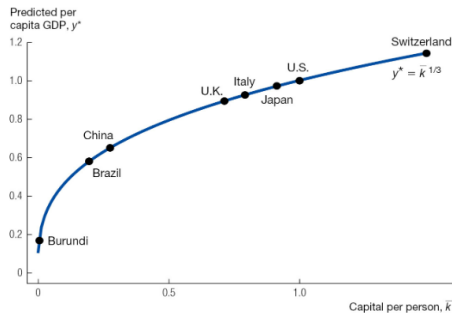


The power of the Cobb Douglas Model

Country	Observed capital per person, $\bar{k}$	Predicted per capita GDP $y = \bar{k}^{-1/3}$	Observed per capita GDP
United States	1.000	1.000	1.000
Switzerland	1.488	1.142	1.203
Japan	0.914	0.971	0.634
Italy	0.792	0.925	0.651
Spain	0.716	0.895	0.645
U.K.	0.714	0.894	0.707
Brazil	0.195	0.580	0.233
China	0.275	0.650	0.226
South Africa	0.165	0.548	0.200
India	0.082	0.434	0.107
Burundi	0.005	0.172	0.013



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India	0.082	0.434	0.107
Burundi	0.005	0.172	0.013





Calculating “A” or TFP

- Our model is inaccurate, as it overestimates the GDP per capita for countries with lower capital per capita
- To compensate, we can use “A” or Total Factor Productivity to “fix” or correct our model

$$\bar{A} = \frac{y^*}{\bar{k}^{-1/3}}.$$



Calculated TFPs:

Country	Per capita GDP (y)	$\bar{k}^{-1/3}$	Implied TFP ($\bar{A}$)
United States	1.000	1.000	1.000
Switzerland	1.203	1.142	1.054
U.K.	0.707	0.894	0.792
Japan	0.634	0.971	0.654
Italy	0.651	0.925	0.703
Spain	0.645	0.895	0.721
Brazil	0.233	0.580	0.402
South Africa	0.200	0.548	0.365
China	0.226	0.650	0.347
India	0.107	0.434	0.247
Burundi	0.013	0.172	0.074



Conclusion of TFP

- What this tells us:
- Differences in TFP are more important than differences in capital per person
- The Cobb Douglas model is imperfect, the output of countries isn't perfectly mapped by capital and labor inputs, and instead relies on a Total Factor Productivity factor as well
- TFP differences aren't perfectly accounted for, but here are some factors as to why we may observe TFP differences:
 - Human Capital, such as educational differences
 - Different technologies and how they are leveraged
 - Institutions, governments, and historical contexts
 - Misallocation of resources

TFP Practice

1. Assuming no differences in productivity (A) across countries, the production model predicts that differences in GDP per person across countries should be (less than / equal to / greater than) differences in capital per person
2. If a country has 10 times the capital per person of another, and A is the same, the production model predicts it should be richer by what proportion?
3. If a country has 10 times the productivity and the same amount of capital per person, the production model predicts that it should be richer by what proportion?
4. Is TFP or differences in capital per person more important in explaining average income differences across countries?

TFP Practice

1. less than, due to diminishing returns on capital
2. 2.2, since $y = k^{\{1/3\}}$ and $10^{\{1/3\}} = 2.2$
3. 10, because $y = Ak^{\{1/3\}}$
4. According to empirical differences, TFP

What does the Cobb Douglas Function Tell us?

- Total output is determined by the total amount of capital and labor available in the economy. We can achieve higher levels of production if we have more capital (machines) or more people, or both
- The equilibrium wage is proportional to output per workers, while the equilibrium return on capital is proportional to output per unit of capital
- We have constant returns to scale if we scale up capital and labor at the same scale. Increasing just capital or just labor can lead us to have diminishing returns to scale
- The economic outcomes of countries does not just rely on their capital and labor stock, but also on other factors, like technologies, policies, and institutions

Problem Set Questions

- (e) [2 points] In Chapter 4, we will see that GDP is modeled well by this Cobb-Douglas production function:

$$Y = A K^{\alpha} L^{1-\alpha} \quad (1.3)$$

where α is a constant number between 0 and 1 that is also capital's share of income. In the textbook, $\alpha = 1/3$, which is an appropriate level of the parameter for U.S. data.

Take the growth rate of both sides of equation (1.3), treating A , K , and L as variables that potentially have growth rates. Simplify and briefly describe your answer:

1. $z = x / y$, then $g_z = g_x - g_y$.
2. If $z = x \times y$, then $g_z = g_x + g_y$.
3. If $z = x^a$, then $g_z = a \times g_x$.

Problem Set Questions

- (f) [2 points] Using your answer above, describe what happens to Y if A is constant while K and L both increase by 2%. Be specific and briefly explain.
- (g) [2 points] Using your answer above, describe what happens to Y if A grows by 3% while K and L both increase by 2%. Be specific and briefly explain.